

CHAPTER THIRTEEN

FORWARD AND FUTURES MARKETS AND THEIR INFLUENCE ON AGRICULTURAL PRICE FORMATION

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13.1. Introduction and learning objectives

Agricultural production is a decentralised activity. Most agricultural goods must therefore be transported to the places where they are processed and consumed (**spatial transformation**). In addition, many products such as grains, oilseeds, potatoes, sugar beets, fruits, and vegetables are harvested seasonally. Other products, such as milk and meat are produced continuously, but even for these products supply and demand typically display seasonal fluctuations (for example due to the seasonal availability of pasture for animals, or special holiday demand such as Thanksgiving turkey). Hence, storage (**temporal transformation**) is required to match agricultural production and consumption over time.

Production, processing and consumption can all take place within a single vertically integrated company. For example, a farmer might produce grain and feeds it to her own pigs, another might grow grapes and produce wine. However, most of the grain, sugar beets, and oilseeds produced in industrialised countries is processed by third-party companies such as oilseed crushers, sugar factories, compound feed mills and flour mills. The resulting products are then in turn either raw materials for companies in the food industry or are sold directly to retailers for sale to consumers.

The farmers, traders, processors and retailers who are part of this food chain coordinate their activities on markets by means of contracts. Contracts allow the market participants to plan their production, input needs, plant capacity

utilisation, storage and transportation needs, quality requirements, cash flows, etc. Contracts are also important for price determination. In particular, they allow market participants to fix the prices at which future transactions will take place. Since future conditions cannot be known with certainty, fixing prices for future transactions via contracts is a means of allocating risk between market participants who have different preferences for risk (**risk transfer**).

In this chapter we discuss two types of contracts, forward contracts and futures contracts, both of which play a key role in the pricing of many agricultural products. In particular, we will explain:

- how forward contracts are defined and how they are used by market participants,
- the advantages and disadvantages of forward contracts for market participants,
- how futures contracts and futures trading work,
- how market participants can hedge risk via futures contracts – so-called hedging,
- how grain traders can implement risk management strategies using futures contracts, and how this can benefit farmers who do not engage in futures trade themselves,
- that speculation plays an essential but also controversial role on futures markets, and
- that futures markets can only function effectively if price determination on futures and spot markets is not distorted by state intervention.

13.2. Forward contracts

When classifying contracts, a basic distinction can be made between those that are fulfilled immediately and those that are fulfilled at a future date. The definition of ‘immediate’ depends on the market in question. If one purchases a downloadable item such as an e-book online, immediate means within a few seconds or minutes. However, if the Kingdom of Saudi Arabia buys a 60,000 ton shipload of barley in a major German port, transport time

dictates that the earliest possible delivery will take 30 days. In international grain trade, 'immediate' by convention refers to the fastest possible delivery of the contracted quantity and quality of grain to the agreed place of delivery by the agreed mode of transport.

In contrast to immediate delivery, **forward contracts** regulate the future delivery or acceptance of goods. Hence, a forward contract can be defined as a contract that is fulfilled later than immediately.

13.2.1. Design of forward contracts

Forward contracts are contracts negotiated individually between contracting parties in which, in addition to the standard contract parameters (such as product, quantity, quality and place of delivery) the date or period of delivery is also specified. The defining characteristic of a forward contract is therefore the future date or period of fulfilment. The price of a forward contract can be determined when the contract is signed, at the time of delivery, or sometime in between. There is also a great deal of flexibility in defining the delivery date or period, depending on the preferences of the contracting parties. To illustrate this, consider the following typical forward contractual arrangements in the grain trade:

1. The simplest form is a contract for fulfilment at a fixed time.
Example: Delivery of 500 tons of feed wheat, 2021 harvest, ciffo/franko¹ Hamburg, fulfilment in the 1st half of May 2022.
2. Forward contracts can also be fulfilled in fixed monthly or semi-monthly instalments over a specified period of time. *Example: 3,000 tons of feed wheat, 2021 harvest, ciffo/franko Hamburg, delivery of 500 tons per month from January to June 2022.*
3. In international grain trade, optional contracts are also common. According to these contracts, the buyer takes delivery of a shipment,

¹ The abbreviation cif (*cost, insurance, freight*) means that the buyer pays for the goods in the port of destination including costs for insurance and freight. If the contracting parties also agree that the buyer pays for the goods after they have been unloaded at the port of destination, the contract contains the clause ciffo (cif + fo for *free out*). In German, the abbreviation franko has the same meaning for transport by truck as ciffo has for transport by ship, i.e. that all costs of transport, insurance and unloading are covered.

for example **fob**² at a specified seaport, but is free to choose the time of delivery within a defined period. Such optional contracts usually contain clauses that specify how much advance notice the buyer must give the seller prior to fulfilment, and what storage charges the buyer incurs per interval of elapsed time within the defined period. These storage charges are called reports and can be defined on a daily, semi-monthly or monthly basis. *Example: Purchase of 60,000 tons of wheat, fob Rostock, 2021 harvest, minimum 12.5% protein, maximum 14% moisture, buyer's option of delivery between February and May 2022, buyer must give 10 days' notice, price + 0.70 €/ton report per half-month.*

13.2.2. Economic motives for conclusion of forward contracts

In the grain trade, forward contracts are the rule and **spot contracts** tend to be used only when farmers sell directly after the harvest. Forward contracts are the central control instrument for sales, production, storage and logistics between the economic actors on grain markets. In addition, they allow actors to fix or hedge price in advance and thus to control revenues (sellers) and costs (buyers).

Processors of agricultural products such as grain and oilseed often sell more or less continuously – e.g. flour mills to retailers and bakers; oilseed crushers to retailers, compound feed and biodiesel producers. Hence, processors require a sufficient supply of raw materials throughout the year. A processor can ensure this supply by own stockholding or by contract suppliers to make continuous deliveries. Own storage makes a processor independent. However, it is expensive to store the amount of raw material required for an entire year of processing, because of the physical costs of storage, and the financial costs of the money tied up in stocks. Furthermore, many processing sites are centrally located on waterways or in large agglomerations. At harvest time, when logistics are tight, it is scarcely possible to transport large quantities of grain from decentralised production centres over several hundreds of kilometres to these concentrated locations. Consequently, most processors only have storage capacities of a few weeks or months of production, and they use forward contracts with suppliers to ensure supply in instalments over the course of a year.

² The abbreviation fob (*free on board*) means that the buyer takes delivery of the goods on board a ship in the specified export harbour, and is responsible for all costs such as insurance and freight incurred thereafter.

Forward contracts for delivery in instalments also provide planning security for the grain suppliers, for example large farms or traders with country elevators who purchase from farms. Forward contracts enable these suppliers to calculate how much storage capacity they require. In addition, grain processors usually buy grain delivered to the plant, i.e. the supplier provides the transport carrier. Forward contracts for delivery in instalments therefore allow for a continuous utilisation of the suppliers' transport carriers - truck, rail or ship. Forward contracts are thus a central element of cost management and the efficient utilisation of storage and processing capacities, and transport infrastructure.

Both forward as well as spot contracts specify all of the parameters that are important for the suppliers and processors. Very large processors usually specify their own individual purchasing conditions. In some markets, standardised forward contracts have also emerged that are used by several processors as a point of reference. Dozens of parameters can be specified in forward contracts. In addition to basic parameters such as the product, quantity, and the date and location of fulfilment, contracts often include detailed clauses on:

- transport carriers; the mechanism for triggering instalments,
- tolerances and associated penalties for over- or under-delivery,
- quality parameters and tolerances as well as surcharges and/or discounts for qualities that do not fall within these tolerances,
- the right to refuse a delivery if certain parameters are not met,
- agreements on the quality management system to be used, on controls, and on the way in which sampling is carried out, and
- agreements on the responsible mediator or jurisdiction in the event of a dispute.

13.2.3. Risks and disadvantages of forward contracts

Forward contracts are basically individually negotiated contracts. It is not easy for either the seller or the buyer to withdraw from a forward contract in the event of a change in the conditions for fulfilment. If a seller is unable to supply as foreseen in a forward contract, he can purchase goods at the same conditions (quality, location, etc.) from a third party to fulfil the

contract. Nevertheless, the original seller remains the buyer's contract partner. When the forward contract is fulfilled, the buyer pays the original seller, who in turn pays the third party from whom he purchased the goods.

Such chain transactions involving standardised contracts for typical qualities of grain and delivery locations are quite common in grain trading. In Germany, for example, standard contracts for milling wheat and feed barley franko/ciffo Hamburg are sometimes sold and resold as often as five or even ten times. The same holds for standardised contracts for wheat, barley or rye franko Südoldenburg in northwest Germany, where there is a concentration of compound feed producers for the local livestock industry. At each link in such a chain, contract conditions and payments have to be controlled and processed, which can be complicated, time-consuming and error-prone.

Forward contracts, like all physical contracts, are associated with the risks of non-performance and non-payment. The contracting parties therefore pay close attention to whom they do business with, and almost all trading houses assess the so-called **counterparty risk** before concluding a contract, i.e. they check the creditworthiness of the contracting party and how well it has executed contracts in the past.

In summary: forward contracts are appropriate instruments for planning and controlling procurement, storage, production, logistics and sales. They also serve to fix prices between contract partners. However, this form of contract is very individual, and only fungible and liquid to a limited extent. Futures contracts can be seen as a reaction to these disadvantages. As we will show in the following sections, futures and forward contracts together provide market participants with highly customised price-fixing and risk-hedging instruments.

13.3. The futures contract

Futures contracts and their predecessors were already being used in the 17th century, e.g. at the Osaka rice exchange in Japan. In Chicago, the Chicago Board of Trade (CBOT) was founded in 1848 and the first forward contract was traded in 1851 (Chicago Board of Trade, 1997). Starting in 1865, the CBOT took steps to standardise these contracts. Such contracts were called **futures contracts** (futures for short). Futures were also traded in Germany, on the Berlin Commodity Exchange. However, futures trade in Germany was stopped in 1886 because Prussian Junkers (a class of aristocrats with

large agricultural estates and considerable political influence) suspected that speculation on the futures market was lowering grain prices. Hence, they lobbied successfully for stricter regulation of exchanges. It would take until 1993 for an amendment to the law to make futures trading possible in Germany again.

13.3.1. The role of futures contracts

Futures contracts can be traded for a variety of different product groups. The *Futures Industry Association* (FIA) collects data from 52 exchanges worldwide and publishes statistics on the trade in futures and options on futures³ for seven product groups: stock indices, stocks, interest rates, currencies, energy (including electricity), agricultural products, non-precious metals and precious metals. According to FIA (2020), almost 34.5 billion contracts were traded worldwide in 2019, including roughly 19 billion futures and 15 billion options contracts. With 12.5 billion contracts, futures on stock indices accounted for around 36% of global trading volume, followed by stocks with 17.6%, interest rates with just under 14%, and currency futures with just over 11%.

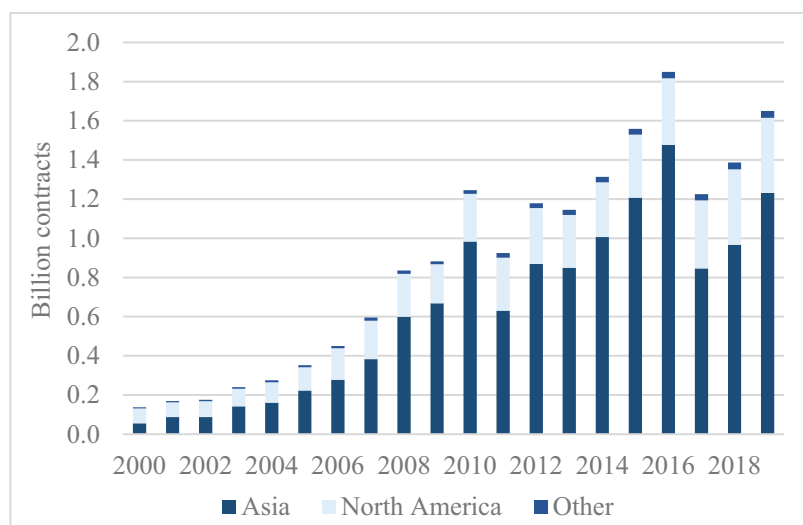
Futures trading in agricultural products has developed rapidly over the past 20 years (Fig. 13.1). According to FIA, only 137 million contracts were traded in 2000, but by 2019 this number had risen to 1.65 billion. North America (especially the US) has accounted for only 20-30% of the number of contracts traded over the last 5 years, compared with 68-80% in Asia (especially China), which has become by far the most important trading centre for futures on agricultural products. In Europe, the number of contracts traded has increased steadily from 4 million in 2000 to 14 million in 2010 and 27 million in 2017. Overall, however, the exchanges in Europe are less important, with a constant share of only about 2% of the global trading volume.

In Europe, futures trading for agricultural products takes place on several commodity futures exchanges. Of particular importance are Euronext (*MATIF - Marché à Terme International de France*, with futures on wheat, rapeseed, rapeseed meal and rapeseed oil, maize and contracts on skimmed milk powder, sweet whey powder, butter and fertilisers), and the Liffe

³ An option is the right to buy or sell something at a later date and at a fixed price. Whereas futures establish obligations, options merely establish rights. Options also play a role in agriculture but are not discussed further here. For more information on options, see e.g. Hull (1998).

(*London International Financial Futures and Options Exchange*), which is part of the NYSE (*New York Stock Exchange*) and on which contracts on sugar and feed wheat are traded.

Figure 13.1 The number of traded futures on agricultural product markets by region of the world



Source: FIA (2020), own illustration.

13.3.2. The nature of futures contracts

Futures contracts are agreements to buy or sell an asset at a specific future settlement date. The detailed contract specifications are standardised and defined by the exchange that provides the contract, e.g. Euronext. The only remaining parameter that the transacting parties must agree on is the price. Due to its great importance for the European grain trade, we present in Table 13.1 the specification of the Euronext MATIF wheat No. 2 futures contract (hereinafter the MATIF wheat contract).

Table 13.1 The specification of the Euronext MATIF wheat No. 2 futures contract

Designation	MATIF-wheat No. 2 futures contract
Contract code	EBM
Trading place	Paris
Clearing	LCH S.A. (a company based in Paris, see below)
Algorithm	The central order book applies a price/time trading algorithm, priority is given to the first order at the best price
Product	Wheat produced in the EU
Moisture	Max. 15%
Broken grains	Max. 4%
Sprouted grains	Max. 2%
Impurities	Max. 2%
Hagberg falling number	Min. 220 seconds
Protein content	Min. 11% in dry matter
Specific weight	Min. 76 kg/hl
Differences between delivered and specified quality	In case of differences, discounts apply in accordance with Incograin No. 23 and Technical Annex 2
Mycotoxins	Applicable EU legislation for unprocessed cereals
Delivery points	In an approved silo in Dunkirk, La Pallice, Montoir, Nantes or Rouen (France).
Delivery months	September, December, March and May
Contract size	50 t
Currency	Euro and Eurocents per t
Tick size	25 Eurocents per t = 12,50 €/contract
Trading times	10:45 – 18:30 (UTC+1)
Last trading day	18:30 (UTC+1) on the tenth calendar day of the delivery month (if no business day, then on the following business day)
Notification day	The first business day following the last trading day
Delivery period	Each business day from the last trading day until the end of the respective delivery month
Delivery limits	Delivery limits are set by LCH S.A. and are published 80 days before the last trading day and are applicable for 12 days before the last trading day of the delivery month

Source: Euronext (2020).

The contract specifications include the trading venue, the responsible clearing institution and a rule for the algorithm of the central order book. We discuss this further below. The contract specifications also include the physical quality parameters described in Table 13.1, such as moisture and protein content, and how differences between these parameters and actual delivered quality will be managed. The specifications also list the possible delivery points, which in the case of the MATIF wheat contract are all accredited silo operations in France. The seller decides at which of the delivery points the delivery will take place. This is called the **seller's option**, and it has important implications for the pricing of a futures contract, as we will explain below in Section 13.3.5.

Based on the specification of the futures contract, the buyer knows exactly which minimum quality he will receive, and the seller knows which minimum quality he must deliver, if the contract is physically delivered. Physical delivery is possible in principle and takes place for all contracts that have not been traded back (**offset** - see below) by the end of their last trading day. The possibility of physically delivering wheat of the quality specified in the futures contract is central to the price formation of the contract. It ensures that on the last trading day of a contract, the price of the contract and the spot market price of wheat (of the quality and at the delivery point specified in the contract) will be equal. On the last trading day, therefore, futures prices and spot prices converge.

All other contract specifications shown in Table 13.1 are of a technical nature. They regulate the contract size (50 t for wheat), the minimum price movement that is traded (0.25 eurocents per t, which corresponds to € 12.50 per 50 t contract), the trading hours, the last trading day and the exact procedures for physical delivery, such as the notification times and the details of where delivery shall take place if the seller exercises the seller's option.

13.3.3. The trading in futures contracts

Since all other parameters of a futures contract are standardised, only the price of the contract remains to be determined. As a rule, futures trading is carried out on behalf of clients by authorised brokerage houses or banks acting as brokers who place buy or sell orders for a specific contract directly in the exchange's trading system. Clients usually place their buy or sell orders with their brokers by telephone or directly via the brokers' electronic trading systems.

The order in which incoming orders are placed on the exchange is regulated by the algorithm described in Table 13.1. In the case of the MATIF wheat contract, for example, this ensures that the sell order for a specific contract and specific price is booked in the same rank order in which it was entered into the exchange's system (first come, first served). If, for example, client X places an order to sell 50 September 2022 wheat contracts, i.e. 2,500 t, at a price of 190 €/t, then this order will enter the queue behind all of the other sell orders for the same contract at the same price. Only when the other contracts ahead in this queue have been purchased will client X's 50 contracts be up for sale. These 50 contracts will then be sold all together or in parts, depending on the number and sizes of the orders in the corresponding queue of buy orders for the same contract at the same price. The exchange's electronic trading system displays the number of sell and buy orders placed for a specific contract. When the electronic trading system has matched client X's sell order with one or more corresponding buy orders, then the seller and the buyer(s) have traded 50 t of wheat as specified in the contract and are informed of this electronically. If this trade is the last one to take place on a given day, the price of 190 €/t will be the official daily closing price for the September 2022 contract on that day. However, exchanges often reserve the right to publish the average price of a number of the last trades as the official closing price.

13.3.4. Market integrity, margin system and the clearing mechanism

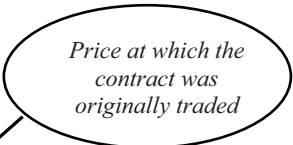
Futures trading is anonymous, i.e. the buyers and sellers of a particular contract have no knowledge of each other's identity. Hence, they cannot assess counterparty risk, as described above for forward contracts. Instead, the integrity of the futures contract, i.e. its unconditional fulfilment and the settlement of profits and losses, is guaranteed by a **clearing mechanism**. On the Euronext, this is regulated by the clearing bank LCH S.A. (see Table 13.1).

The clearing mechanism is based on a system of margins. Margins are amounts of money that are deposited in a trustee account with the clearing bank. The clearing bank acts as a third party to the trade between a buyer and a seller and requires that both deposit **margins**, which are either a percentage of the contract value or a fixed amount per contract. When a contract is traded, both the buyer and the seller must first deposit the so-called **initial margin**, which is an amount at least large enough to cover the maximum loss that could be incurred on the next trading day. This

maximum loss is determined by the trading limit of the futures contract that is set by the exchange, i.e. maximum amount by which the price of a contract might fluctuate within a certain period of time, beyond which trading for this futures contract will be temporarily suspended.

The buyer of a futures contract suffers a loss if the market price for this contract falls. This is because by purchasing the contract, the buyer has committed to pay a certain price in return for the quantity, quality, time and place of delivery specified in the contract. If the price of the contract falls, then the same quantity and quality at the same time and place could be purchased for less. Consider the example shown in Table 13.2. The futures contract was originally traded for 190 €/t, and in the meantime its market price has fallen to 185 €/t. The initial margin of 5 €/t covers the buyer's loss of 5 €/t exactly. The clearing bank then demands an additional 5 €/t margin from the buyer, the so-called **margin call**. By this mechanism, the buyer of the contract is obliged to always deposit with the clearing bank an amount equal to the maximum loss that he/she might incur in the course of the next day's trading.

Table 13.2 An example of the initial margin and margin calls in €/t for different price developments

Buyer			Euronext	Seller		
Exchange demands	Sum margin	Margin	Price	Margin	Sum margin	Exchange demands
Margin call	€ 35	€ 5	€ 160			
Margin call	€ 30	€ 5	€ 165			
Margin call	€ 25	€ 5	€ 170			
Margin call	€ 20	€ 5	€ 175			
Margin call	€ 15	€ 5	€ 180			
Margin call	€ 10	€ 5	€ 185			
Initial margin	€ 5	€ 5	€ 190	€ 5	€ 5	Initial margin
			€ 195	€ 5	€ 10	Margin call
			€ 200	€ 5	€ 15	Margin call
			€ 205	€ 5	€ 20	Margin call
			€ 210	€ 5	€ 25	Margin call
			€ 215	€ 5	€ 30	Margin call
			€ 220	€ 5	€ 35	Margin call

The same logic applies for the seller, who suffers a loss when the price of the contract increases. The seller has committed to deliver the quantity and quality at the time and place of delivery specified in the contract, in return for a certain price. If the price of the contract increases, then the same quantity and quality, at the same time and place, could be sold for more.

Trading in futures contracts can thus bind large amounts of capital in the form of deposited margins. In the event of sharp price hikes, market participants who have sold futures contracts have to inject a lot of capital in the form of margin calls, and may run into liquidity problems as a result. This occurred in 2007/08, for example, or more recently following Russia's attack on Ukraine in February 2022, when prices for many agricultural commodities rose sharply.

13.3.5. Obligations of the market participants

If a contract is traded at time t_0 , the seller (A) undertakes to deliver the commodity and quality underlying the future at the specified delivery time and place. The seller's position is called **short**.⁴ The buyer of a future (B), on the other hand, undertakes to accept delivery of the commodity in accordance with the contract specifications. The buyer's position is called **long**. Market participant A can cancel his/her delivery obligation (**short position**) in t_1 without affecting B if he/she can find another market participant (C) who is willing to assume this delivery obligation in return for a price. The result would be that C now has a short position and B continues to have a **long position** on the commodity futures market. Participant A would no longer have a position the commodity futures market because his/her initial short position has been **offset**. Of course, B can also offset his/her long position by transferring the associated obligation to accept delivery to another market participant.

On a commodity futures exchange, the margins that the seller and buyer have deposited are matched and balanced by the clearing bank. In the example in Table 13.2, if the price of the contract has fallen to 170 €/t by t_1 , the buyer realises a loss of 20 €/t when he/she offsets his/her long position by selling the contract. The clearing bank will therefore retain 20 of the total 25 €/t of margin that the buyer has deposited since t_0 , and use this money to cover the profits that the seller has realised since t_0 . Thus, commodities do not necessarily have to be physically exchanged as a result of trade in futures contracts. Sellers and buyers can always eliminate their

⁴ A helpful mnemonic: 'short' and 'sell' both begin with the letter 's'.

obligations to make/take delivery by offsetting their positions, with the clearing house balancing any accumulated profits and losses. In practice, the share of contracts that are physically delivered is usually less than 1% of the contract volume.

Table 13.3 summarises and contrasts the main characteristics of forward and futures contracts.

Table 13.3 Characteristics of forward and futures contracts

Characteristic	Futures contracts	Forward contracts
Specification	Quantity, quality, and place and time of delivery are standardised	Individually negotiable
Liquidity	Correlated with trading volume on the futures exchange	Must be negotiated individually, not guaranteed. Fungibility is guaranteed only on a few paper markets
Integrity	Guaranteed by the exchange via the clearing bank	Counterparty risk must be managed by the respective contract parties
Payment method	Continuous by means of the margin mechanism	Individual, mostly as a lump sum at or after delivery
Delivery	As a rule, as contracts are offset therefore no delivery	Physical, countertrade or loss compensation
Transaction costs	Broker fees and interest on deposited margin	Negotiation and settlement costs

Source: Nelson (1985, p. 17).

13.3.6. The pricing of futures contracts

In the following section we explain the pricing of futures contracts and how futures prices are linked to spot market prices by the so-called **basis**. The basis is the difference between the current or future spot price at a certain location, and the price of a certain futures contract. A distinction is made between the **forward basis**, which is the difference between the spot price and the forward price ($p_{t_0}^S - p_{t_0 \rightarrow t_1}^{FUT}$) at a certain point in time t_0 , and the **maturity basis**, which is the difference between the spot price and the futures price when the futures contract matures ($p_{t_1}^S - p_{t_1 \rightarrow t_1}^{FUT}$), i.e. at the

time of delivery. In these equations, $p_{t_i \rightarrow t_j}^{FUT}$ denotes the price at time t_i of the futures contract with maturity t_j , and $p_{t_k}^S$ denotes the spot price at time t_k .

Arbitrage affects the basis and thus the relationship between futures and spot prices. For example, if the futures price is higher than the spot price for a commodity with identical quality characteristics at time t_1 , then arbitrageurs will sell futures contracts at price $p_{t_1 \rightarrow t_1}^{FUT}$, purchase equivalent amounts of the commodity in question on the spot market at the lower price $p_{t_1}^S$, and physically deliver the futures contracts. By doing so, the arbitrageurs will tend to drive down the price of the futures contract, and drive up the spot price. As a result, arbitrage will drive the maturity basis to zero. However, the maturity basis will be smaller (larger) than zero if the commodity traded on the spot market is of lower (higher) quality than that specified in the futures contract.

Based on analogous considerations, we expect that in a perfect market futures and forward prices will be almost identical. If, for example, the forward price $p_{t_0 \rightarrow t_1}^{FWD}$ exceeds the futures price $p_{t_0 \rightarrow t_1}^{FUT}$ at t_0 , arbitrageurs will sell commodities forward and buy contracts on the futures market with the intention of insisting on physical delivery of the futures contracts at time t_1 when they mature. With the delivered commodity they can fulfil the forward contract and thus realise the profit $p_{t_0 \rightarrow t_1}^{FWD} - p_{t_0 \rightarrow t_1}^{FUT}$. Consequently, the prices of forward contracts will generally be derived from futures prices, with adjustments for quality differences. Later in this chapter we show how this relationship affects price determination in the grain trade.

Abstracting from quality differences and different delivery locations, it can be assumed that the forward basis will increase linearly with increasing time horizon in the course of a crop year. Formally, the relationship between a futures price $p_{t_0 \rightarrow t_1}^{FUT}$ and the spot price $p_{t_0}^S$ can be represented as follows:

$$p_{t_0 \rightarrow t_1}^{FUT} = p_{t_0}^S (1 + r_{t_0 \rightarrow t_1}) + L_{t_0 \rightarrow t_1} \quad (1)$$

where

$r_{t_0 \rightarrow t_1}$ = the opportunity cost of an amount of capital equal to the spot price over the period from t_0 to t_1 ,

and

$L_{t_0 \rightarrow t_1}$ = the cost of storage per unit of the good from t_0 to t_1 .

If equation (1) does not hold exactly, arbitrageurs will be able to realise certain profits. If, for example $p_{t_0 \rightarrow t_1}^{FUT} > p_{t_0}^S(1 + r_{t_0 \rightarrow t_1}) + L_{t_0 \rightarrow t_1}$ they will buy the commodity on the spot market for $p_{t_0}^K$ and at the same time take a corresponding short position on the futures market (i.e. sell a futures contract for $p_{t_0 \rightarrow t_1}^{FUT}$). When the futures contract matures, they can deliver the commodity that they purchased on the spot market, and the difference between $p_{t_0 \rightarrow t_1}^{FUT}$ and $p_{t_0}^K$ will be more than enough to cover the cost of purchasing the commodity and storing it in the interim.

13.3.7. The seller's option

As we explained above, the seller's option allows the seller to select the delivery point from a specified list. We illustrate this using the example of the MATIF rapeseed contract, which specifies eight delivery points (in France, Belleville, Metz and Frouard on the Moselle River; in Germany, Bülstringen, Vahldorf and Magdeburg on the *Mittellandkanal*, and Würzburg on the Main River; and in Belgium, Ghent).

For the delivery of contracts that have not been offset prior to maturity, sellers will choose a delivery point where they can deliver the rapeseed at the lowest cost. Until 2012, for example, there was a surplus of rapeseed in the new *Länder* of the former German Democratic Republic. This rapeseed had to be transported via the *Mittellandkanal* to oil mills on the Lower Rhine, which cost roughly 12 €/t. The price fob barge *Mittellandkanal* was thus 12 €/t below the price cif on the Lower Rhine. At that time, rapeseed could be transported the Moselle to the Lower Rhine for 6 €/t, so that the price fob Moselle was generally 6 €/t below cif Lower Rhine. Similarly transport costs from the port of Ghent to the Lower Rhine were about 5 €/t, so fob Ghent was correspondingly 5 €/t below cif Lower Rhine. The price at the *Mittellandkanal* was always the lowest price derived from the price cif Lower Rhine and thus represented the most favourable option for sellers who had to deliver. This was anticipated by the market participants. As a result, the maturity basis for the price fob *Mittellandkanal* always amounted to 0 €/t, the MATIF rapeseed contract reflected fob *Mittellandkanal*, and most deliveries took place on the *Mittellandkanal*.

Since then, however, rapeseed processing capacities in the new *Länder* have grown and rapeseed production has declined. As a result, the region has become a net importer of rapeseed. At the same time, the EU as a whole has also developed from a net exporter to a net importer of rapeseed. The lowest price is now usually the cif Ghent price, which in turn represents the

cheapest delivery option. The MATIF rapeseed contract therefore now reflects the Ghent location. Oil mills located on the *Mittellandkanal* must now pay a basis (traders refer to it as a ‘premium’) of up to 14 €/t on top of the MATIF rapeseed price, to cover transport costs from the port of Ghent.

13.4. Risk management with futures markets

13.4.1. Basics of hedging on futures markets

When market participants use futures markets to protect themselves against the risk of price fluctuations for physical goods, they are said to be **hedging**, in the sense of ‘fencing in’ or ‘limiting’. In the following we will explain how a hedge functions, what risks are associated with hedging, and what role speculation plays on futures markets.

In general, a distinction is made between **long** and **short hedges**. A seller who would like to fix the price of a future sale of physical goods on the spot market in order to protect against falling prices, is called a short hedger. A farmer, for example, could try to fix the selling price for his/her next harvest by means of a short hedge. If, on the other hand, a buyer attempts to fix the price of a future purchase of physical goods to protect against rising prices, we refer to a long hedge. Typical long hedgers are processors of agricultural products, such as flour mills, that try to lock in future raw material costs.

As an example of a long (short) hedge, assume that a trader intends to buy (sell) physical goods on the spot market at price $p_{t_1}^S$ at time t_1 . To do this, he sells (buys) a forward contract at time t_0 and price $p_{t_0 \rightarrow t_1}^{FWD}$. Without hedging, there is a price risk, since $p_{t_1}^S$ is unknown at time t_0 . If $p_{t_1}^S$ turns out to be lower than the forward price, the trader has lost, because he/she could have paid less for the goods in question. This risk can be reduced (hedged) by a countertrade on the futures market. For this purpose, the trader buys (sells) futures contracts at price $p_{t_0 \rightarrow t_1}^{FUT}$ and time t_0 , and thus enters into a long (short) position on the futures market. The purchase of the futures contracts can be viewed as a temporary substitute for the transaction to be carried out later on the spot market. The long (short) futures position is offset by selling (buying) futures contracts at time t_1 and price $p_{t_1 \rightarrow t_1}^{FUT}$. The profits or losses from the transactions on the futures market are offset by the clearing house as explained above and either credited or charged to the trader. At the same time, the trader acquires (sells) the physical commodity on the spot market.

Formally, this long hedge can be represented as follows:

$$\begin{aligned}\pi_H &= qp_{t_0 \rightarrow t_1}^{FWD} - qp_{t_1}^S - qp_{t_0 \rightarrow t_1}^{FUT} + qp_{t_1 \rightarrow t_1}^{FUT} \\ &= q(p_{t_0 \rightarrow t_1}^{FWD} - p_{t_1}^S) - q(p_{t_0 \rightarrow t_1}^{FUT} - p_{t_1 \rightarrow t_1}^{FUT})\end{aligned}\quad (2)$$

where

π_H = the profit after the long hedge,

q = the quantity traded,

$p_{t_0 \rightarrow t_1}^{FWD}$ = the forward price in t_0 for delivery in t_1 ,

$p_{t_1}^S$ = the spot price in t_1 ,

$p_{t_0 \rightarrow t_1}^{FUT}$ = the futures price in t_0 for delivery in t_1 , and

$p_{t_1 \rightarrow t_1}^{FUT}$ = the futures price in t_1 for delivery in t_1 .

Thus, the profit of a long hedge π_H equals the change in the value of the quantity traded on the spot market ($p_{t_0 \rightarrow t_1}^{FWD} - p_{t_1}^S$) minus the change in the value of the quantity traded on the futures market ($p_{t_0 \rightarrow t_1}^{FUT} - p_{t_1 \rightarrow t_1}^{FUT}$). If these changes mirror one another, price risk is reduced. This will be the case if both $p_{t_0 \rightarrow t_1}^{FUT}$ and $p_{t_0 \rightarrow t_1}^{FWD}$, and $p_{t_1 \rightarrow t_1}^{FUT}$ and $p_{t_1}^S$ move in parallel. As explained above, arbitrage will ensure that these conditions are met. First, $p_{t_0 \rightarrow t_1}^{FWD}$ and $p_{t_0 \rightarrow t_1}^{FUT}$ will move in parallel as forward prices are derived from futures prices, with allowance for quality differences between the commodity traded in the forward and futures markets. Second, $p_{t_1 \rightarrow t_1}^{FUT}$ and $p_{t_1}^S$ will also move in parallel as the maturity basis tends towards zero, again with allowance for quality differences. Hence, any loss (gain) on the spot market will be compensated by a corresponding gain (loss) on the futures market - i.e. $(p_{t_0 \rightarrow t_1}^{FWD} - p_{t_1}^S) \approx (p_{t_0 \rightarrow t_1}^{FUT} - p_{t_1 \rightarrow t_1}^{FUT})$.

These relationships are illustrated in Table 13.4. A trader buys wheat for 184 €/t on November 28, 2019 (whether on the spot market or on the forward market is irrelevant for the example). At the same time, he sells a March 2020 wheat futures contract (i.e. enters into a short position) on the MATIF at a price of 180 €/t. The basis for this contract is therefore 4 €/t. At this point, the trader has bought (is long on) the physical commodity, and

has sold a futures contract (is short on the futures market). This position (physical commodity long, futures short) is called **premium long**. The opposite position, in which a trader has sold the physical commodity and hedged the price by buying a futures contract (physical commodity short, futures long) is called **premium short**.

On February 15, 2020, the trader sells the physical commodity for 180 €/t, which is 4 €/t below the November 2019 purchase price. At the same time, the trader offsets the futures contract, i.e. he takes the opposite position and buys back the futures contract, which has also fallen by 4 €/t. Since the prices on the futures market and the spot market have moved in parallel, as described above (both have fallen by 4 €/t), the basis has not changed. Without a hedge, the trader would have lost 4 €/t on the physical transaction; however, this loss is completely offset by a profit on the futures market. The result is therefore 0 €/t and the hedge is perfect.

Table 13.4 Position premium long: prices fall, basis remains unchanged

	Physical		MATIF		Basis	Result
	Buy	Sell	Buy	Sell		
28 Nov 2019	€ 184			€ 180	€ 4	
15 Feb 2020		€ 180	€ 176		€ 4	
Profit/loss		€ -4		€ 4	€ 0	€ 0

Source: Own illustration.

The same transaction has a different outcome for the trader if the basis increases. This is the case when the physical value falls less (or increases more) than the price on the MATIF. In Table 13.5, the physical price falls by 4 €/t as in the previous example, but the MATIF price falls by 6 €/t. Hence, the basis increases from 4 to 6 €/t. The trader again loses 4 €/t on the physical transaction, but in this case this loss is more than compensated by the gain of 6 €/t from the short hedge on the futures market.

Table 13.5 Position premium long: prices fall, basis increases

	Physical		MATIF		Basis	Result
	Buy	Sell	Buy	Sell		
28 Nov 2019	€ 184			€ 180	€ 4	
15 Feb 2020		€ 180	€ 174		€ 6	
Profit/loss		€ -4		€ 6		€ 2

Source: Own illustration.

Similarly, if the physical price falls more (or increases less) than the price on the MATIF, the trader with the long premium position incurs a loss (Table 13.6). In this case, the loss on the physical transaction is not fully offset by a profit from the short hedge on the futures market. As we will explain in greater detail in the next section, the changes in the basis that we have assumed in the last two examples (Tables 13.5 and 13.6) can be triggered by factors such as volatile transport costs or weather-related changes in the qualities available on the spot market.

Table 13.6 Position premium long: prices fall, basis falls

	Physical		MATIF		Basis	Result
	Buy	Sell	Buy	Sell		
28 Nov 2019	€ 184			€ 180	€ 4	
15 Feb 2020		€ 180	€ 178		€ 2	
Profit/loss		€ -4		€ 2		€ -2

Source: Own illustration.

We present a final example in Table 13.7 to show that a falling basis leads to a profit if the trader is premium short instead of premium long. In this case, the physical commodity is first sold (physical commodity short) and the price is hedged by buying a futures contract (futures long). A falling basis then leads to a profit, since the profit on the physical transaction is greater than the loss from the long hedge on the futures market. Conversely, premium short leads to losses if the basis increases.

Table 13.1 Position premium short: prices fall, basis falls

	Physical		MATIF		Basis	Result
	Buy	Sell	Buy	Sell		
28 Nov 2019		€ 184	€ 180		€ 4	
15 Feb 2020	€ 180			€ 178	€ 2	
Profit/loss		€ 4		€ -2		€ 2

Source: Own illustration.

10.4.1.1. Risks associated with hedging

Hedging is often compared to insurance, but this comparison is flawed. The insurance principle is based on spreading possible losses over many policyholders. In contrast, hedging means offsetting one risk by means of an opposite transaction so that the combined risk of both transactions together is smaller than the risk of one alone. However, as the examples above have illustrated, risk is not completely eliminated. Since the basis can increase or fall, the basis risk always remains.

We introduced the concepts of basis risk and margin risk above. Further risks are the liquidity risk and the standard quantity risk, both of which can also affect the profitability of a hedge. In the following, we review each of these types of risk in turn.

The price risk in spot market transactions can only be reduced by hedging if spot and futures market prices move largely in parallel and the basis is therefore roughly constant over time. In reality, however, this is rarely the case. **Basis risk** arises because the basis in t_1 is unknown and cannot be forecast exactly at time t_0 . In fact, a hedge does not completely eliminate price risk on the spot market, it merely exchanges this price risk for basis risk. In the case of a long hedge, a basis that increases over time ($[p_{t_0 \rightarrow t_1}^{FUT} - p_{t_1}^K] > [p_{t_1 \rightarrow t_1}^{FUT} - p_{t_1}^K]$) leads to a loss, while a basis that falls over time ($[p_{t_0 \rightarrow t_1}^{FUT} - p_{t_1}^K] < [p_{t_1 \rightarrow t_1}^{FUT} - p_{t_1}^K]$) leads to a gain. In the case of a short hedge, the opposite effect occurs in each case.

Changes in basis can be caused by a variety of factors. These include volatile transport costs. If, for example, freight rates increase or important waterways become temporarily impassable due to weather conditions, transport costs will increase. This will affect the costs of physically

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delivering a futures contract from other locations (in space). As a result, the difference between the futures price and the spot price at another location will increase. Unless transport costs are constant, spot prices are decoupled from the futures prices to a certain extent, resulting in basis fluctuations.

Changes in the relative qualities of the goods on the spot and futures markets can also generate basis fluctuations. While the MATIF wheat contract provides for a minimum protein content of 11% (see Table 13.1), on average over the years, German milling wheat has a roughly 1% higher protein content, which leads to a positive basis. However, if extreme weather in Germany leads to quality losses in a particular year, the basis can weaken.⁵

In most markets, however, the correlation between futures and spot prices is sufficiently high (i.e. basis risk is sufficiently low) to ensure that hedging usually leads to a reduction in the price risk of spot market transactions. Indeed, it is sometimes the case that the correlation between the futures price for one product and the spot price for a different but related product is sufficiently high to allow successful hedging. For example, rapeseed and soybean prices are often so positively correlated that transactions on the spot market for rapeseed can be hedged with soybean futures. In this case, we speak of a **cross hedge**.

Liquidity risk, which has been a problem for example on European commodity futures markets for dairy or meat products, is closely related to basis. A futures market is liquid when a sufficient number of market participants (including speculators⁶) trade on that market and hedgers are therefore able to trade and, in particular, offset any and all open contract volumes at the prevailing market price at any time. In a market with low liquidity, i.e. a low trading volume, it can be difficult to offset an open position because – as explained above – every transaction on a futures market involves matching, i.e. the simultaneous presence of a buyer and a seller who are both willing to trade at a certain price. On a market with low liquidity, hedgers might have to make price concessions to offset large positions, and this can lead to changes in the basis.

Whereas liquidity risk is a characteristic of a particular futures market, **margin risk** can create liquidity problems for individual farmers. As we

⁵ Prehn et al. (2016) and Vollmer et al. (2020) analyse the determinants of the basis between spot prices for wheat in Germany and the price of the MATIF wheat contract.

⁶ The role of speculators on futures markets is discussed in Section 13.4.2.

explained above, a farmer who wants to hedge by selling futures contracts for wheat (short hedge) will have to make margin calls on an ongoing basis if prices rise sharply. Depending on the financial situation of the farm and the size of the short position, this can lead to considerable liquidity problems for a farm (see case study 1 below).

Standard quantity risk is a problem that can arise when a hedger only produces or purchases small quantities. A futures contract specifies a certain quantity of for example 50 tons in the case of the MATIF wheat contract. It is not possible to purchase or sell fraction of a contract. If a farmer who expects to harvest 25 tons of wheat attempts to hedge by selling a futures contract for 50 tons, he/she will be speculating on the futures market with the 25-ton difference. Similarly, a farmer who hedges an expected harvest of 70 tons with a single 50-ton contract will be speculating with 20 tonnes on the spot market.

10.4.1.2. Determining the optimal hedge ratio

A market participant who hedges all of his/her expected spot market transactions on the futures market ton-for-ton is said to have a hedge ratio of 1, or a 'full hedge'. The hedge ratio is defined as the ratio of the quantity that is hedged on the futures market to the quantity that (will be) traded on the spot market.

$$H_R = \frac{q_H}{q_S} \quad (3)$$

where

H_R = the hedge ratio,

q_H = the quantity hedged on the futures market, and

q_S = the quantity traded on the spot market.

A hedge ratio of 1 means that the hedger does not take on any standard quantity risk. However, a hedge ratio of 1 is not necessarily optimal; depending on the hedger's individual risk affinity/aversion and the level of risk associated with a hedge, the optimal hedge ratio can deviate from 1, sometimes considerably. The term 'under hedge' ($H_R < 1$) is used to describe the situation in which a market participant hedges only a part of the quantity traded on the spot market. Similarly, if a larger quantity is

hedged on the futures than on the spot market we refer to an ‘over hedge’ ($H_R > 1$). A market participant who hedges nothing ($H_R = 0$) is, by definition, a **speculator** who bears the price risk on the entire volume of his/her open spot market position.

What is a market participant’s optimal hedge ratio? First of all, it should be clear that even an infinitely risk-averse market participant will not choose a full hedge. This is due to the different types of risk discussed in the previous section, and in particular the basis risk. A risk-averse market participant is interested in minimising overall risk, not only price risk. The overall risk consists of the price risk on the spot market and the basis risk on the futures market. To illustrate this, we begin with equation (2)⁷

$$\pi_H = q(p_{t_0 \rightarrow t_1}^{FWD} - p_{t_1}^S) - q(p_{t_0 \rightarrow t_1}^{FUT} - qp_{t_1 \rightarrow t_1}^{FUT})$$

which can be modified as follows:

$$\pi_H = q_S p_{t_0 \rightarrow t_1}^{FWD} - q_S p_{t_1}^S - q_H p_{t_0 \rightarrow t_1}^{FUT} + q_H p_{t_1 \rightarrow t_1}^{FUT} \quad (4)$$

where q_S and q_H are the spot and hedged quantities as defined above. The risk-averse hedger is interested in minimising the variance of π_H , defined as:

$$Var(\pi_H) = E(\pi_H - E(\pi_H))^2 \quad (5)$$

where

$Var(\pi_H)$ = the variance of the profit after a hedge (π_H), and

$E(\pi_H)$ = the expected value of π_H .⁸

Substituting equation (4) into equation (5) leads to:

$$Var(\pi_H) = E([q_S(p_{t_0 \rightarrow t_1}^{FWD} - p_{t_1}^S) - q_H(p_{t_0 \rightarrow t_1}^{FUT} - p_{t_1 \rightarrow t_1}^{FUT})] - E[q_S(p_{t_0 \rightarrow t_1}^{FWD} - p_{t_1}^S) - q_H(p_{t_0 \rightarrow t_1}^{FUT} - p_{t_1 \rightarrow t_1}^{FUT})])^2 \quad (6)$$

⁷ See Malliaris (1999, pp. 40-42).

⁸ Equation (2) (or 4) applies to the case of a long hedge. In the case of a short hedge, all of the terms in this equation have the opposite sign. However, since these terms are all squared in the derivation of the optimal hedge ratio (below), the derivation applies to both long and short hedges.

Since at time t_0 both the quantity traded on the spot market (q_S) and the two prices $p_{t_0 \rightarrow t_1}^{FUT}$ and $p_{t_0 \rightarrow t_1}^{FWD}$ are known, equation (6) can be modified to:

$$\begin{aligned} Var(\pi_H) &= E \left([q_S(p_{t_0 \rightarrow t_1}^{FWD} - p_{t_1}^S) - q_H(p_{t_0 \rightarrow t_1}^{FUT} - p_{t_1 \rightarrow t_1}^{FUT})] \right. \\ &\quad \left. - [q_S(p_{t_0 \rightarrow t_1}^{FWD} - E(p_{t_1}^S)) \right. \\ &\quad \left. - q_H(p_{t_0 \rightarrow t_1}^{FUT} - E(p_{t_1 \rightarrow t_1}^{FUT}))] \right)^2 \\ &= E \left(q_S(E(p_{t_1}^S) - p_{t_1}^S) - q_H(E(p_{t_1 \rightarrow t_1}^{FUT}) - p_{t_1 \rightarrow t_1}^{FUT}) \right)^2 \end{aligned} \quad (7)$$

We next expand equation (7) to obtain:

$$\begin{aligned} Var(\pi_H) &= E \left(q_S^2 (p_{t_1}^S - E(p_{t_1}^S))^2 \right. \\ &\quad \left. - 2q_S q_H (p_{t_1 \rightarrow t_1}^{FUT} - E(p_{t_1 \rightarrow t_1}^{FUT})) (p_{t_1}^S - E(p_{t_1}^S)) \right. \\ &\quad \left. + q_H^2 (p_{t_1 \rightarrow t_1}^{FUT} - E(p_{t_1 \rightarrow t_1}^{FUT}))^2 \right) \\ &= q_S^2 Var(p_{t_1}^S) - 2q_S q_H Cov(p_{t_1}^S, p_{t_1 \rightarrow t_1}^{FUT}) + q_H^2 Var(p_{t_1 \rightarrow t_1}^{FUT}) \end{aligned} \quad (8)$$

where

$Cov(p_{t_1}^S, p_{t_1 \rightarrow t_1}^{FUT})$ = the covariance between $p_{t_1}^S$ and $p_{t_1 \rightarrow t_1}^{FUT}$.

Equation (8) can be further modified as follows:

$$Var(\pi_H) = q_S^2 \left(Var(p_{t_1}^S) - 2 \frac{q_H}{q_S} Cov(p_{t_1}^S, p_{t_1 \rightarrow t_1}^{FUT}) + \frac{q_H^2}{q_S^2} Var(p_{t_1 \rightarrow t_1}^{FUT}) \right) \quad (9)$$

Next, we substitute equation (3) ($H_R = \frac{q_H}{q_S}$) into equation (9):

$$\begin{aligned} Var(\pi_H) &= q_S^2 \left(Var(p_{t_1}^S) - 2H_R Cov(p_{t_1}^S, p_{t_1 \rightarrow t_1}^{FUT}) \right. \\ &\quad \left. + H_R^2 Var(p_{t_1 \rightarrow t_1}^{FUT}) \right) \end{aligned} \quad (10)$$

The first-order condition for minimising the variance of π_H by choosing the optimal hedge ratio H_R^* is:

$$\frac{\partial Var(\pi_H)}{\partial H_R} = q_S^2 \left(-2Cov(p_{t_1}^S, p_{t_1 \rightarrow t_1}^{FUT}) + 2H_R Var(p_{t_1 \rightarrow t_1}^{FUT}) \right) = 0 \quad (11)$$

Solving equation (11) results in:⁹

$$H_R^* = \frac{\text{Cov}(p_{t_1}^S, p_{t_1 \rightarrow t_1}^{FUT})}{\text{Var}(p_{t_1 \rightarrow t_1}^{FUT})} \quad (12)$$

If there is no basis risk, the prices on the spot market and on the futures market are perfectly correlated. The variance of the futures price (in the denominator) and the covariance of the futures and the spot prices (in the numerator) will therefore be identical, and a full hedge will be optimal. As the basis risk increases, the covariance of the futures and spot prices decreases, and with it the optimal hedge ratio. In the extreme case, both the covariance of the futures and spot prices and the optimal hedge ratio equal zero.

An estimate of the optimal hedge ratio can be derived from the linear regression equation of spot prices on futures prices at time t_1 .

$$p_{t_1}^S = \beta_0 + \beta_1 p_{t_1 \rightarrow t_1}^{FUT} \quad (13)$$

The least squares estimate of the slope coefficient β_1 in equation (13) is:

$$\widehat{\beta}_1 = \frac{\sum(p_{t_1}^S - \overline{p_{t_1}^S})(p_{t_1 \rightarrow t_1}^{FUT} - \overline{p_{t_1 \rightarrow t_1}^{FUT}})}{\sum(p_{t_1 \rightarrow t_1}^{FUT} - \overline{p_{t_1 \rightarrow t_1}^{FUT}})^2} \quad (14)$$

Dividing the numerator and the denominator of equation (14) by $(n - 1)$ (where n = the number of observations), we obtain:

$$\widehat{\beta}_1 = \frac{\sum(p_{t_1}^S - \overline{p_{t_1}^S})(p_{t_1 \rightarrow t_1}^{FUT} - \overline{p_{t_1 \rightarrow t_1}^{FUT}})/(n-1)}{\sum(p_{t_1 \rightarrow t_1}^{FUT} - \overline{p_{t_1 \rightarrow t_1}^{FUT}})^2/(n-1)} = \frac{\text{Cov}(p_{t_1}^S, p_{t_1 \rightarrow t_1}^{FUT})}{\text{Var}(p_{t_1 \rightarrow t_1}^{FUT})} = H_R^* \quad (15)$$

Thus, the optimal hedge ratio H_R^* can be derived as the slope of a linear regression equation of spot prices on futures prices in t_1 ($H_R^* = \widehat{\beta}_1$).

⁹ Since the second derivative $\frac{\partial^2 \text{Var}(\pi_H)}{\partial H_R^2} = 2q_S^2 \text{Var}(p_{t_1 \rightarrow t_1}^{FUT})$ is always positive, equation (12) minimises $\text{Var}(\pi_H)$.

Other methods of deriving the optimal hedge ratio besides simple least squares regression have been proposed in the literature. For example, error correction models (introduced in Chapter 10) can also be used. Both least squares and error correction models generate constant values of H_R^* that do not vary over time. Using so-called GARCH models, it is also possible to estimate time-varying hedge ratios that account for the possibility that price variances and covariances might change over time.¹⁰

Case study 1: Optimal hedge ratios for wheat producers in Germany

Vollmer and von Cramon-Taubadel (2020) empirically analyze the potential of price hedging using the MATIF wheat contract as a risk management tool for farmers in Germany. Among other things, the authors calculate optimal hedge ratios using a simple least squares regression with the spot price of wheat FOB Rostock as the dependent variable (p_t^S) and the price of the MATIF wheat contract as the independent variable (p_t^{FUT}). They use weekly price data from September 2001 to April 2016. They also assume two short hedging strategies: first, that the farmer hedges her crop 12 months in advance directly after sowing, and second, that she waits until the spring and hedges her harvest 6 months in advance. The latter strategy might make more sense, as the farmer can better estimate the harvest volume to be hedged in the spring, after the crops have emerged from the winter months. Vollmer and von Cramon-Taubadel (2020, Table 2) present the following regression results (p-values in brackets):

$p_t^S = < 0.001 + 0.455(p_{t+12Months}^{FUT})$	$p_t^S = < 0.001 + 0.375(p_{t+6Months}^{FUT})$
(0.001) (0.035)	(0.001) (0.028)

Accordingly, the estimated optimal hedge ratio is 45.5% (37.5%) for the strategy with a time horizon of 12 (6) months. These values are very similar to those estimated using error correction models instead of least squares regression. The authors also estimate time-varying optimal hedge ratios using GARCH models, which will not be discussed further here.

Vollmer and von Cramon-Taubadel (2020, Table 10) also simulate the liquidity risks associated with the hedging strategies mentioned. Their results show that especially in years that were characterised by high swings in wheat prices (e.g. 2006/07 and 2009/10), large margin calls of up to

¹⁰ GARCH models are beyond the scope of this chapter. An overview of different methods for calculating optimal constant or dynamic hedge ratios is provided by Chen et al. (2003).

€ 6,000 per contract could arise. A farmer who sold 10 contracts to hedge 500 t of wheat, for example, would therefore have to have the financial means to make margin calls of up to € 60,000.

13.4.2. Hedging vs. speculation

So far, we have reviewed how reciprocal transactions on spot and futures markets are used by market participants to reduce price risk. However, futures contracts can also be traded by market participants who have no interest in the physical commodity, i.e. who speculate. Speculators are the third (and often the largest) group of participants in futures markets, along with arbitrageurs and hedgers. Speculators trade on futures markets, buying or selling contracts and later offsetting them, without owning or otherwise intending to deal in the corresponding physical commodity.

A distinction can be made between **short speculators** (also called bearish speculators) and **long speculators** (bullish speculators). A short speculator expects prices to fall over time and therefore goes short by selling at t_0 on the futures market for time t_1 . He thus enters into a delivery obligation without owning the commodity. At t_1 he offsets his futures position by buying and realises a profit if the price of the contract has fallen since t_0 . In contrast, a long speculator bets on rising prices in the future. She goes long in the futures market at t_0 for time t_1 and thus enters into a purchase obligation. At t_1 she offsets her futures position by selling and realises a profit if the price of the contract has increased since t_0 . The short speculator's profit expectation is therefore

$$\pi_S = q(p_{t_0 \rightarrow t_1}^{FUT} - p_{t_1 \rightarrow t_1}^{FUT}) \quad (16)$$

and the long speculator's profit expectation is

$$\pi_L = q(-(p_{t_0 \rightarrow t_1}^{FUT}) + p_{t_1 \rightarrow t_1}^{FUT}) \quad (17)$$

with

q = the quantity traded on the futures market.

We illustrate these relationships with an example of short speculation in Table 13.8. Assume that a speculator expects wheat prices to fall between November and March. In order to profit from this expected price decline,

the speculator goes short on the futures market. She is not using the expected co-movement of futures and cash prices to hedge physical goods; instead she is deliberately speculating that falling prices will lead to a profit on the futures market. This example makes clear that speculation does not minimise but rather entails additional risks.

In scenario 1, prices fall as the trader expected, from 184 €/t in November to 180 €/t in the following March. The trader thus makes a profit of 4 €/t from the futures transaction. In scenario 2, the price rises to 190 €/t and the speculator suffers a loss of 6 €/t.

Table 13.2 An example of short speculation

		Futures price per ton	
		Scenario 1	Scenario 2
November (t_0)	Sale of one March wheat futures contract	184	184
March (t_1)	Purchase of one March wheat futures contract	180	190
Profit on the futures market		4	-6

Source: Own illustration.

A speculator considers whether she believes that the price at t_1 quoted on the futures market at t_0 is correct, or whether she believes that the price will change between t_0 and t_1 . For this purpose, she collects and analyses information on the market conditions that will determine the price of the futures contract at t_1 . If her analysis leads to a (subjective) price expectation $p_{t_1 \rightarrow t_1}^{FUT}$ that differs from the current price $p_{t_0 \rightarrow t_1}^{FUT}$, she will take a long or short speculative position depending on the sign of the difference. The speculator's profit, if any is realised, can be interpreted as a reward for the successful collection and analysis of relevant market information. Since false price expectations lead to losses, speculators must take great care when collecting and analysing market information.

The futures price for an agricultural product such as wheat thus reflects all the market information available to all of the market participants at a given time. New information is continuously being generated by factors such as weather developments, exchange rate fluctuations, trade reports, policy changes, epidemics such as African Swine Fever or the coronavirus, and disruptions due to conflict such as Russia's attack on Ukraine. This information is transmitted onto futures markets and thus into futures prices as it arises, often on a minute-by-minute basis.

The speed with which new information is priced into futures markets has increased significantly as electronic trading platforms have replaced the floor trading that was common in the past. At the same time, the importance of so-called algorithmic or automated trading has grown. Algorithmic trading is based on computer programs that are designed to anticipate market developments by evaluating past price movements, the news situation, the weather and developments on other markets worldwide. Estimates suggest that by 2019, algorithmic trading accounted for 50-70% of global agricultural futures trade. At the Chicago Mercantile Exchange, for example, automated computer transactions now account for over 50% of the total trading volume in corn and soybeans futures, and over 60% for wheat and live cattle (Haynes and Roberts 2019, Table 3). Because of their comparatively low transaction costs and high speed of information processing, futures markets often react more rapidly and strongly to new information than spot markets. Futures markets therefore act as a highly efficient and fast form of information exchange.

While speculation is a vital source of information on futures markets, and contributes essentially to their liquidity, it is also controversial. The so-called agricultural price crisis of 2007/08 led to renewed accusations that speculators deliberately manipulate and increase prices for staple foods such as wheat in order to make profits at the expense of the hungry. If these accusations could be substantiated, stronger regulation or even bans on futures trading might be justified. However, there is little sound theoretical or empirical evidence that speculators systematically use futures markets to increase staple food prices.¹¹

An important distinction can be made between informed and uninformed speculators. Informed speculators have information about so-called **fundamentals**, i.e. factors that influence demand and supply and thus determine the price level of a good. Informed speculators assume that they have an information advantage, i.e. that they have more, better or earlier information, or that they are able to assess the implications of new information for future pricing earlier or better than other market participants. They use this advantage to form expectations about future prices and take corresponding long or short positions. As explained above,

¹¹ Recall (see section 10.3 above) that futures markets were banned in the 19th century in Germany because producers argued that speculation was lowering grain prices.

their futures trading ensures that futures prices always reflect all available market information.

Uninformed speculators do not base their futures positions on fundamentals. Instead they analyse past price movements for patterns such as trends. If they are convinced, for example, that a sustained rising trend has set in, they take long positions in the expectation of further rising prices. If many speculators follow such a strategy (herd behaviour), their expectations of rising prices can become self-fulfilling and a futures price can move away from its fundamental value, at least temporarily. Algorithmic trading might reinforce such price movements, which are sometimes called price or speculative **bubbles**. A speculative bubble must burst sooner or later: if a futures price has moved away from its fundamental value, informed speculators and other market participants can take counter-positions and make safe profits. To do this, however, they must be able to pick the right time when the herd behaviour will subside and prices will return to fundamental values.¹²

The literature contains some empirical evidence for the existence of price bubbles in staple food futures markets (see case study 2). However, the literature is not unanimous, and there is no convincing evidence to support the claim that the 2007/08 agricultural price crisis was largely or even exclusively triggered by speculation. On the contrary, the 2007/08 price peak can be explained with reference to market fundamentals. The USDA (the US Department of Agriculture) publishes estimates of the global supply situation - i.e. stocks, production, trade and consumption - for all major global agricultural markets in its monthly World Agricultural Supply and Demand Estimates (WASDE) reports (USDA, 2020). These reports represent the most extensive and consistent source of publicly available information on agricultural markets, and they have a correspondingly large influence on markets and prices. However, they are not infallible. The February 2008 WASDE report described an historically tight supply situation on world grain markets, and grain prices peaked after it was released. When the USDA later revised its assessment and released less pessimistic numbers, grain prices quickly fell. Was the price peak thus the result of the USDA's initial misjudgement, or was it caused by speculative market participants who based their trades on the USDA assessments?

¹² As Gilbert (2010, p. 4) aptly writes: "...there is no easier way to lose money than to be right, but to be right too soon".

To prove the existence of a price bubble on a futures market, one must prove that the prices on this market have moved away from their fundamental values. Thus, one must have complete information on all relevant fundamentals and how they influence prices. In practice, however, it is not possible to categorically rule out that an important fundamental factor has been overlooked. When Robles et al. (2009, p. 2), referring to the 2007/08 price crisis, claim that "changes in supply and demand fundamentals cannot fully explain the recent drastic increase in food prices", they are implicitly assuming that they have complete empirical information on all relevant market fundamentals and their effects on price levels – certainly a heroic assumption.

Case study 2: Speculation with agricultural commodities

Adämmer and Bohl (2014) look for evidence of speculative bubbles on American futures markets for maize, soybeans and wheat. To this end, they analyse the relationship between agricultural prices, the price of crude oil and an index of various exchange rates using monthly data from 1993 to 2012. Crude oil prices and exchange rates are considered to be fundamental factors that influence the prices of agricultural commodities. It is expected that rising crude oil prices will lead to increasing agricultural prices, while rising real exchange rates will lead to a decline in demand and thus to falling prices. If rapid increases in agricultural prices are followed by rapid falls, and these peaks are not observed in crude oil prices or exchange rates, this is considered to be evidence of a speculative bubble.

Adämmer and Bohl's (2014) results provide evidence for the existence of speculative bubbles in the wheat futures market. This confirms the findings of earlier studies such as Gilbert (2010) and Gutierrez (2013). For the soybean and maize markets, Adämmer and Bohl (2014) do not find clear evidence that speculation influences price developments.

Another important challenge for studies that analyse the influence of speculation on futures prices lies in the need to categorise market participants into speculators and hedgers. In theory, the speculative motive can be clearly distinguished from the hedging motive, but in practice many participants cannot be unambiguously assigned to one group or the other. A trader can simultaneously hedge and speculate on futures markets, and any hedge that deviates from the optimal hedge ratio contains an element of speculation. An extensive literature has looked specifically at the activity and effect of so-called hedge funds, which clearly act as speculators on agricultural futures markets. But even this literature does not offer firm and

robust evidence of a strong influence on futures prices (see e.g. Sanders and Irwin 2010).

Note that farmers often engage in speculative behaviour on agricultural markets. A farmer who does not use forward or futures contracts to hedge his future harvests is automatically in a long position. If a poor harvest leads to expectations of increasing prices, farmers will often speculate by holding back grain for later sale. If many farmers do this, it will reduce supply and drive up prices.

Since there is at least some evidence for the existence of price bubbles on agricultural futures markets, could government regulation of these markets improve their performance and ensure that they do not distort price formation? Yes, perhaps, but not necessarily. Market failure is not a sufficient condition for government intervention, because any such intervention also carries the risk of policy failure. Careful analysis must first consider whether there are feasible government interventions whose costs do not exceed the cost of the market failure that they are supposed to correct. The activity of speculators is essential for the functioning of futures markets as institutions that serve price discovery and that allow for the exchange of risk between market participants with different willingness and ability to bear risk. The call for regulation to curb excessive speculation is understandable, but if such regulation is not able to clearly distinguish and target ‘excessive’ regulation, it will generate more costs than benefits. Speculation in staple food markets has been debated heatedly for many decades, and it will likely remain a controversial topic in future scientific and political discourse.

13.4.3. Futures markets as reference markets for the pricing of spot and forward contracts

Although farmers in Germany use futures markets as an important source of information, they rarely actively participate in these markets. In a survey conducted by Adämmer et al. (2014), only 10% of farmers reported hedging on commodity futures markets. Even in the US, which is home to many of the largest commodity futures exchanges and where there is a long, unbroken tradition of futures trading, most farmers do not participate - surveys often report shares below 10% (see e.g. Carter 1999, p. 216).

Why is this the case? Possible explanations include small farm structures and the high costs of opening a futures trading account, i.e. high transaction

costs per hedged contract.¹³ Other explanations are the liquidity risk of possible margin calls (see case study 1 above) as well as the information costs of futures trading for farmers who are already very busy with day-t-day farming operations. Pannell et al. (2008) present simulation results showing that hedging only generates notable benefits for farmers with above-average risk aversion. Farmers in the EU receive secure per hectare direct payments that stabilise their income and thus possibly increase their willingness to take risks when marketing their products, i.e. reduce their demand for risk reduction through hedging.

Another important explanation lies in the fact that many farmers do not need to participate on futures markets directly because they can do so indirectly via forward contracts that are offered to them by traders who do make active use of futures markets. When farms wish to hedge future harvests, they can sign forward contracts with local traders who, in turn, hedge their forward transactions with farmers on futures markets. On Euronext exchange, for example, contracts can be traded up to three years in advance of the delivery date; the May 2025 MATIF wheat contract, for example, has been trading since May 2022. Usually, the volume of trade for a new contract is initially low, but trade picks up over time, and a contract generally becomes sufficiently liquid to permit hedging within 18 to 24 months of its delivery date.

An example of this indirect or derived participation by farmers on futures markets is shown in Table 13.9. In February 2020, a farmer decides to hedge a certain quantity of her upcoming 2020 wheat harvest. Because she has some on-farm storage capacity, she decides to sell forward for October 2020, a few months after the harvest. On February 18, the MATIF wheat contract for December 2020 is quoted at 189.25 €/t. The farmer's local trader expects that it will realise a basis of +3.00 €/t fob Hamburg in December 2020, i.e. a price of $(189.25 + 3.00) \text{ €/t} = 192.25 \text{ €/t}$. The forward price that the traders is prepared to offer the farmer is derived from this price, minus the transport costs from the farm to (the seagoing vessel in) Hamburg. If, for example, handling and loading in Hamburg costs 5.00 €/t and truck transport from the farmer's storage location to Hamburg costs 7.00 €/t, then the trader can offer the farmer a forward contract for October 2020 delivery for a farm gate price of $(192.25 - 5.00 - 7.00) \text{ €/t} = 180.25 \text{ €/t}$.

¹³ See e.g. Salhofer and Zoll (2005).

Table 13.3 Derived price hedging through agricultural trade

	Physical		MATIF		Basis	Result
	Buy	Sell	Buy	Sell		
18 Feb 2020	€ 192.25*			€ 189.25	€ 3	
20 Oct 2020		€ 182	€ 181		€ 1	
Profit/loss		€ -10.25		€ -2	€ -2	€ -2

** Farm-gate price 180.25 €/t = (192.25 fob Hamburg less handling 5 and truck transport 7) €/t.*
Source: Own illustration.

The trader thus offers the farmer a price that is the futures price plus or minus the expected basis and corrected for location (i.e. transport and handling charges). To hedge this transaction with the farmer, the trader (who is now in a long position) assumes a short position on the futures market by selling a corresponding number of December 2020 MATIF wheat contracts. The liquidity of the futures market (including the activity of speculators) has put the trader in a position to offer the farmer a forward contract and, by extension, price risk reduction.

To complete the example, assume that prices fall between February and October, so that the trader can only resell the wheat he collects from the farmer for a price of 182 €/t fob Hamburg (rather than the expected 192.25 €/t). The trader therefore suffers a loss of 10.25 €/t on the physical market. When the forward contract with the farmer is delivered (October 2020), the trader offsets his short position on the futures market by buying back the corresponding MATIF wheat contracts. The futures price has fallen in parallel with the price on the physical market, but only by 8.25 €/t. The basis is therefore only 1.00 €/t rather than the expected 3.00 €/t (i.e. the basis has fallen by 2.00 €/t), so the gain of 8.25 €/t on the futures market cannot fully compensate for the loss on the physical market.

Overall, in this example, the trader suffers a loss of 2 €/t because the basis has fallen between February when the forward contract was signed, and October when it was delivered. However, the key point in the example is that hedging on the futures market enables the trader to provide derived price risk reduction in the form of forward contracts to many farmers with smaller volumes of production who might not be willing or able to engage in futures trade themselves.

13.5. Preconditions for the functioning of commodity futures markets

As shown in Section 13.3.1, futures markets exist for a whole range of product groups including cereals, oilseeds, oils, sugar, milk and its processed products, orange juice, tea and piglets. This disproves an assumption often made in the past, according to which futures markets can only be derived for storable and standardisable agricultural products. Milk, for example, is not storable and live animals cannot be standardised. In fact, a futures contract can be designed for any product for which a price can be defined at a precise time and place. Even if no physical delivery is possible - as is the case for futures contracts on shipping routes, for example - the futures can be traded on the basis of so-called cash settlements. In this case, the profit and loss between the future's short and long traders is balanced at the time of settlement on the basis of a transparent and clearly documented price.¹⁴

The decisive factor for the success of a futures market is whether it is accepted by market participants and, especially in the case of agricultural products, allows participants to hedge prices. There have been many failed attempts to establish futures markets.¹⁵ In Germany, for example, a commodity futures exchange (the WTB) was launched in Hanover in April 1998. Besides potatoes and pork, a wheat futures market was also established on the WTB, but later abandoned. This was probably not due to the design of the WTB wheat contract, but rather to competition from the MATIF in France. For German grain traders, the correlation between futures prices on the MATIF and grain prices in Germany is sufficient for successful hedging. Under these conditions, when a liquid contract already provides sufficient opportunities for hedgers, it is difficult to establish a new contract.

In general, the attractiveness of a futures contract for a commodity will increase with the variability and unpredictability of the commodity's price. The size of the underlying market, and thus the potential volume of futures trade, also influences the success or failure of a contract. The number of possible trading partners for a contract can also play a role. The larger the number of potential participants, the less influence individual traders will

¹⁴ For a more detailed analysis of the role of storability in price discovery on futures markets, see Yang et al. (2001).

¹⁵ Till (2015) reviews a large number of examples of futures markets that did or did not succeed.

have on price behaviour and the more attractive the contract. The attractiveness of a future also depends on the alternative price hedging options. For example, there are many different futures contracts for wheat traded on exchanges all over the world, but to the best of our knowledge there is no futures contract for flour. This is probably because flour mills, large bakeries and retailers are generally able to manage price risk using contracts.

Finally, the successful establishment of a futures market also depends on the institutional framework in the country where it is traded. First of all, the integrity of the market must be guaranteed. Trading partners must be absolutely certain that the clearing bank will provide a secure and transparent exchange of profits and losses. In addition, market participants who hedge price risks must be absolutely certain that there will be no government intervention on the market in question. In Ukraine and Russia, for example, there have been several plans and attempts to establish commodity futures markets for wheat and other agricultural commodities. However, since politicians have repeatedly intervened in the markets in a discretionary manner, for example by imposing export quotas or even embargoes, hedging is associated with an enormous basis risk. Consider a trader who holds a physical long position inside Russia, and hedges this with a future based on a fob location, for example at a Black Sea port. If an export restriction is imposed, then the trader's physical long will fall in price, while his futures short will rise. Both ends of the trader's hedge are therefore negatively affected by the government's export restriction. Hedging via futures therefore only works if price formation can take place free of state intervention as a result of the interplay of market supply and demand.

13.6. Keywords

- Basis, maturity basis
- Basis risk
- Bubble
- ciffo
- Clearing mechanism
- Counterparty risk
- Cross hedge
- fob
- Forward contract
- Fundamentals
- Futures contract

- Hedge, long hedge, short hedge, hedge ratio
- Liquidity risk
- Long, long position
- Margin, initial margin, margin call
- Margin risk
- Offset
- Premium long, premium short
- Risk transfer
- Seller's option
- Short, short position
- Spatial transformation, temporal transformation
- Speculator, long speculator, short speculator
- Standard quantity risk
- Spot contract

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Exercises

Consider the following products:

- A: Milling wheat on the spot market
- B: Euronext MATIF Wheat No. 2 Futures Contract (delivery May 2020)
- C: CBOT Wheat Futures Contract (delivery May 2020)

The following series of transactions is carried out by a local trader:

1. January 15, 2020:
Buy A: 1000 t for 193 €/t
Sell B: 500 t for 195 €/t
Sell C: 500 t for 192 €/t
2. February 15, 2020:
Buy B: 300 t for 194 €/t
Sell A: 300 t for 190 €/t
3. March 15, 2020
Buy C: 500 t for 170 €/t
Sell A: 500 t for 171 €/t
4. April 15, 2020
Buy B: 200 t for 193 €/t
Sell A: 200 t for 191 €/t

As of April 15, 2020, all positions have been offset.

- a. Calculate the trader's profit or loss from this series of transactions.
- b. What would have happened if the land trader had not hedged, but

rather simply bought 1,000 t of A on the spot market on January 15 and sold this amount again on April 15?

- c. Assess the overall success of the trader's hedge. Was it worthwhile for the trader? What considerations go into your answer?